

Does Liver Radiodensity Predict Chemotherapy Toxicity?

Time-to-event analysis across CAP, OXA, and CAPOX

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Study Objective

- Assess whether liver radiodensity (HU) predicts time to dose-limiting toxicity
- Outcome: number of cycles until toxicity occurs
- Adjusted for patient characteristics and hospital differences

Key takeaway: Liver HU is evaluated for its added predictive value in clinical decision-making.

Toxicity Endpoints Analyzed Separately

- CAP toxicity
- OXA toxicity
- CAPOX toxicity

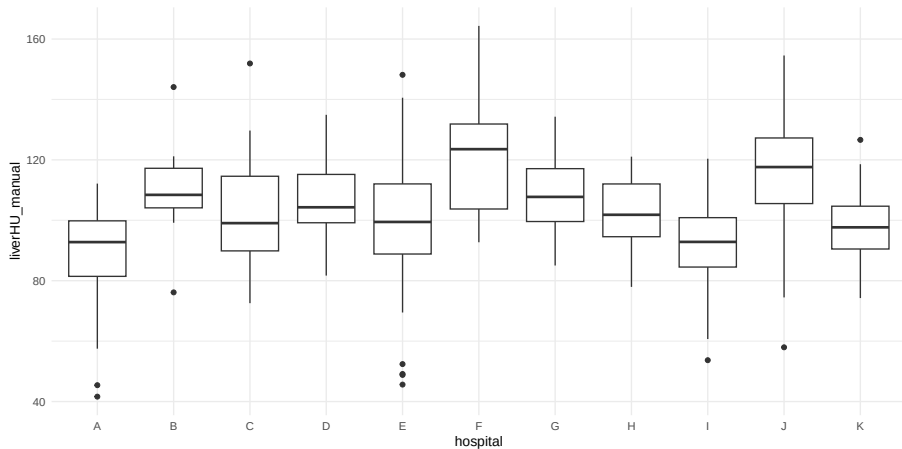
Key takeaway: Each treatment is evaluated using a separate time-to-event model.

Observed Toxicity Rates Differ by Treatment

Endpoint	N	Events	Censored
CAP	344	181	163
OXA	311	254	57
CAPOX	359	257	102

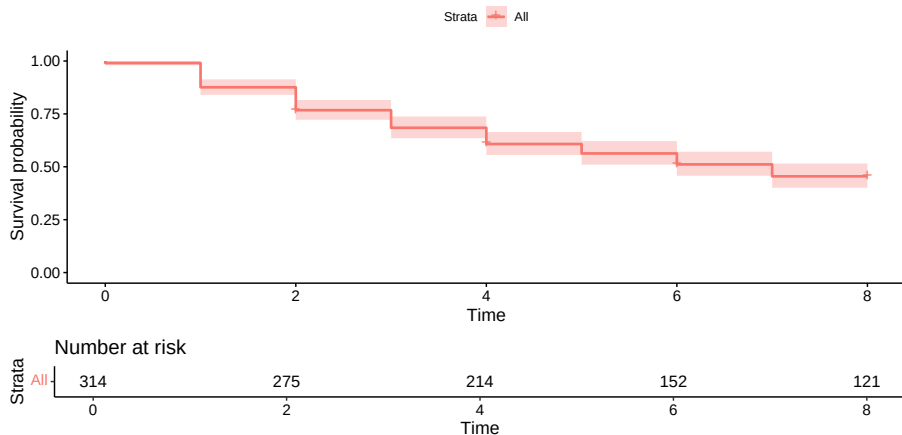
Key takeaway: Event rates differ across treatments, with sufficient events to support survival analysis.

Liver Radiodensity Varies Across Hospitals



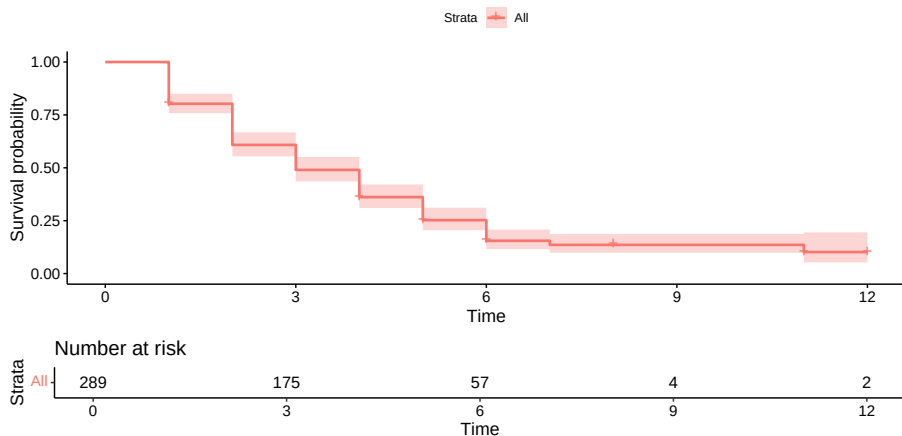
Key takeaway: Liver HU distributions differ between hospitals, suggesting baseline differences that are accounted for through stratification.

Time to Toxicity: CAP



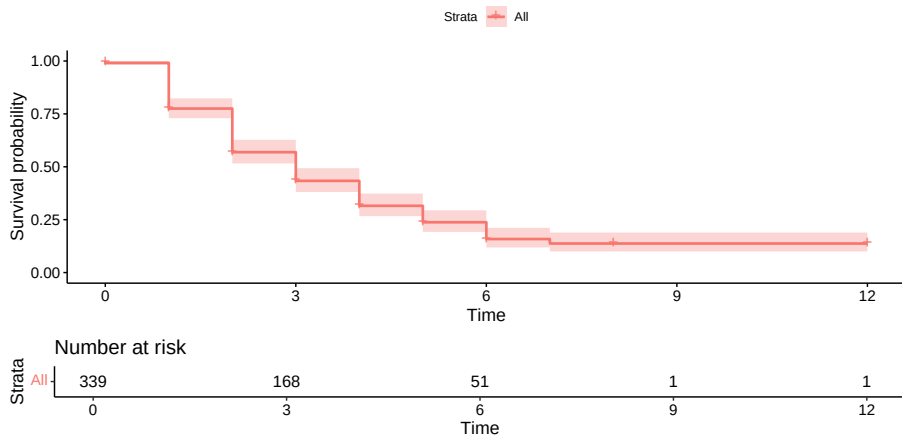
Key takeaway: The probability of remaining toxicity-free declines over cycles, with most events occurring early in treatment.

Time to Toxicity: OXA



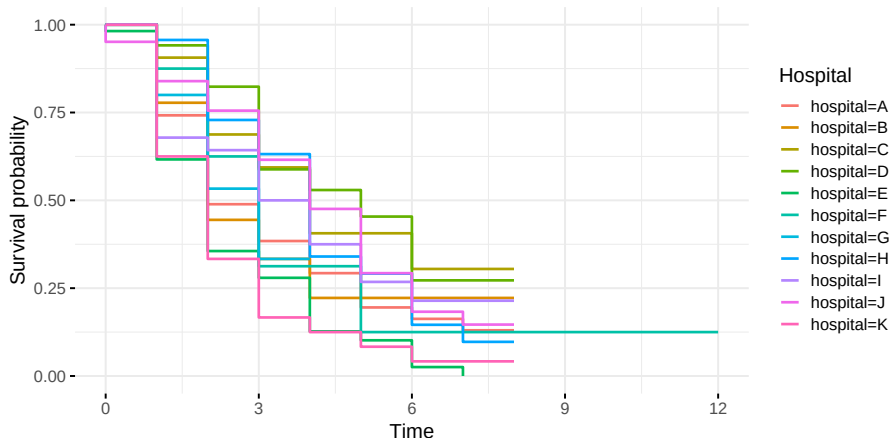
Key takeaway: Toxicity risk is highest in early cycles. Estimates in later cycles are less reliable due to few patients remaining at risk.

Time to Toxicity: CAPOX



Key takeaway: A similar pattern is observed, with most toxicities occurring early and increasing uncertainty in later cycles.

Kaplan-Meier curves by hospital (capox as example)



- toxicity risk varies across hospitals
- differences likely reflect patient mix and clinical practice

Why Use Cox Models?

- Kaplan-Meijer curves are unadjusted and do not account for hospital differences
- Cox models are stratified by hospital
- This allows each hospital to have its own baseline hazard

Why this matters:

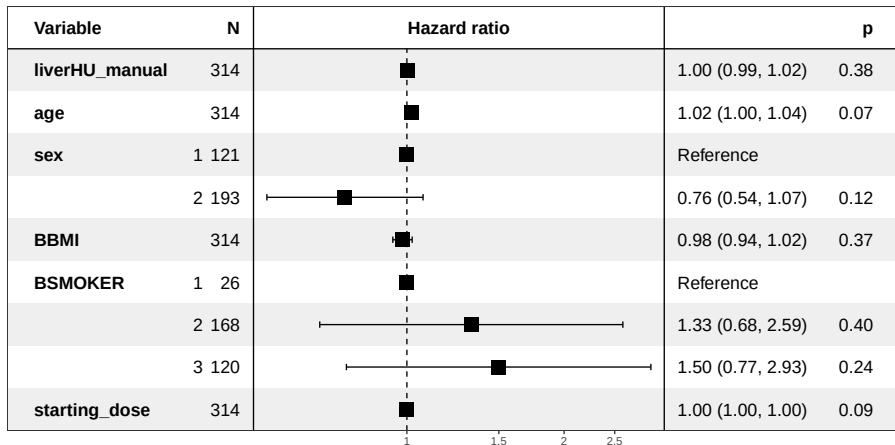
- Accounts for differences in patient populations, practices, and measurement
- Ensures estimated effects (e.g., liver HU) are not confounded by hospital differences

Key takeaway: Results reflect within-hospital comparisons, not differences between hospitals.

Association Between Liver HU and Toxicity

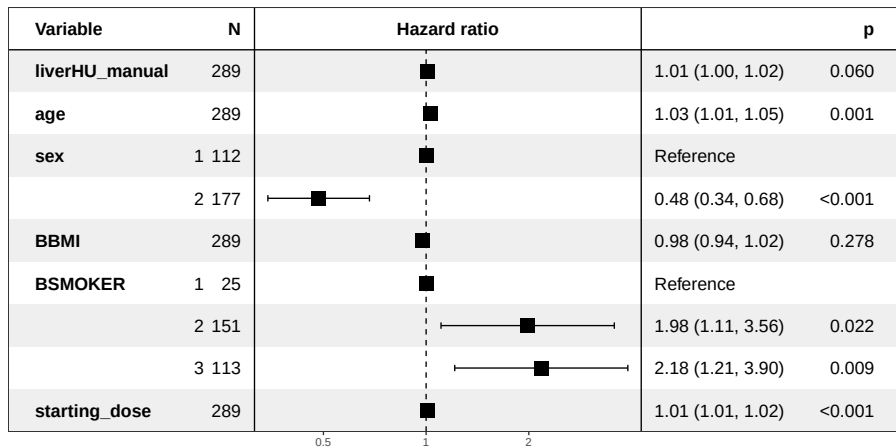
term	estimate	std.error	statistic	p.value	conf.low	conf.high	Model
liverHU_manual	1.00	0.01	0.88	0.38	0.99	1.02	CAP
age	1.02	0.01	1.79	0.07	1.00	1.04	CAP
sex2	0.76	0.18	-1.55	0.12	0.54	1.07	CAP
BBMI	0.98	0.02	-0.89	0.37	0.94	1.02	CAP
BSMOKER2	1.33	0.34	0.83	0.40	0.68	2.59	CAP
BSMOKER3	1.50	0.34	1.18	0.24	0.77	2.93	CAP
starting_dose	1.00	0.00	1.70	0.09	1.00	1.00	CAP
liverHU_manual	1.01	0.00	1.88	0.06	1.00	1.02	OXA
age	1.03	0.01	3.18	0.00	1.01	1.05	OXA
sex2	0.48	0.17	-4.15	0.00	0.34	0.68	OXA
BBMI	0.98	0.02	-1.09	0.28	0.94	1.02	OXA
BSMOKER2	1.98	0.30	2.30	0.02	1.11	3.56	OXA
BSMOKER3	2.18	0.30	2.61	0.01	1.21	3.90	OXA
starting_dose	1.01	0.00	3.59	0.00	1.01	1.02	OXA
liverHU_manual	1.01	0.00	1.22	0.22	1.00	1.02	CAPOX
age	1.01	0.01	0.90	0.37	0.99	1.02	CAPOX
sex2	0.75	0.14	-1.99	0.05	0.57	1.00	CAPOX
BBMI	0.99	0.02	-0.59	0.55	0.96	1.02	CAPOX
BSMOKER2	1.78	0.27	2.14	0.03	1.05	3.01	CAPOX
BSMOKER3	1.83	0.27	2.24	0.03	1.08	3.12	CAPOX

CAP: No Meaningful Association with Liver HU



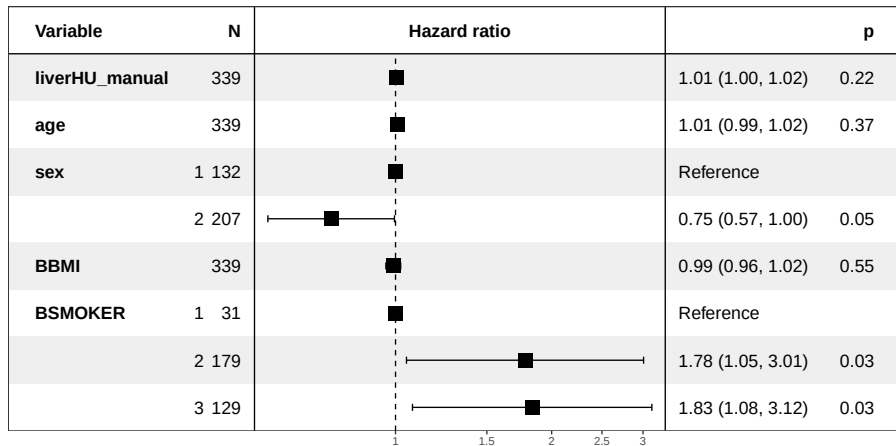
Key takeaway: The estimated effect of liver HU is close to null with no statistically significant association.

OXA: Small Trend, Not Statistically significant



Key takeaway: A slight association is observed, but the effect size is small and not significant

CAPOX: No Consistent Association Observed



Key takeaway: Estimates are close to null, with no clear evidence of an association.

Model Assumptions Hold

- No evidence of violation of the proportional hazards assumption
- Hazard ratios appear stable over time

	chisq	df	p
liverHU_manual	0.2255962	1	0.6348086
age	0.2120542	1	0.6451621
sex	0.7255615	1	0.3943257
BBMI	0.7826862	1	0.3763209
BSMOKER	0.1230178	2	0.9403446
GLOBAL	2.0444171	6	0.9155685

Key takeaway: The Cox model assumptions are satisfied, supporting the validity of the results.

Overall Interpretation

- Liver radiodensity shows no consistent or clinically meaningful association with time to toxicity
- Effect sizes are small and not statistically significant across endpoints
- Other clinical variables (e.g., age, smoking, dose) show stronger associations

Key takeaway: Liver radiodensity does not appear to add meaningful predictive value beyond standard clinical variables even after accounting for hospital effects

No clear association was observed when liver radiodensity was modeled continuously across its full range. A possible follow-up analysis would be to assess whether toxicity risk differs only at clinically meaningful thresholds or extreme radiodensity ranges.